

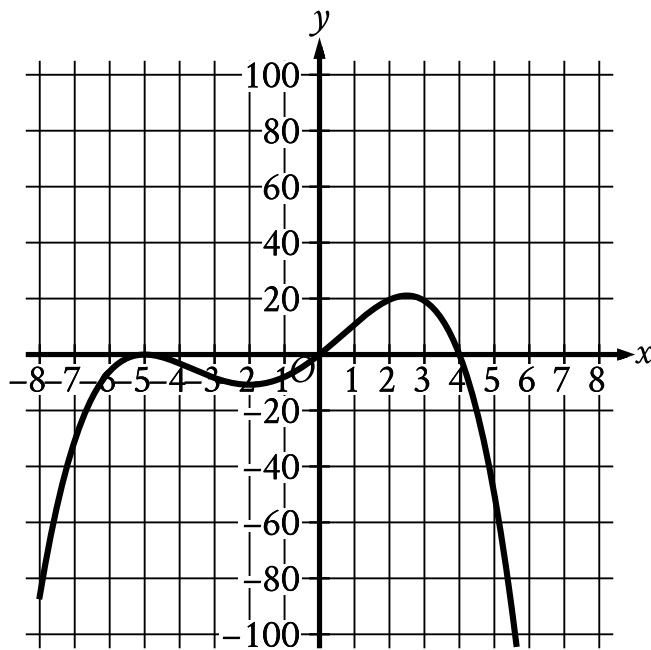
● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

02



- In quadrant 3:
 - The curve rises sharply to touch the x axis at point (negative 5 comma 0).
 - The curve falls gradually to a relative minimum at point (negative 2 comma negative 11).
 - The curve rises gradually to cross both axes at the origin.
- In quadrant 1:
 - The curve rises gradually to a relative maximum at point (2.5 comma 21).
 - The curve falls sharply to cross the x axis at point (4 comma 0).
- In quadrant 4 the curve falls sharply.

Which of the following could be the equation of the graph shown in the xy -plane?

A. $y = -\frac{1}{10}xx - 4x + 5$

B. $y = -\frac{1}{10}xx - 4x + 5^2$

C. $y = -\frac{1}{10}xx - 5x + 4$

D. $y = -\frac{1}{10}xx - 5^2x + 4$

Q2

ADVANCED MATH

A certain college had 3,000 students enrolled in 2015. The college predicts that after 2015, the number of students enrolled each year will be 2% less than the number of students enrolled the year before. Which of the following functions models the relationship between the number of students enrolled, $f(x)$, and the number of years after 2015, x ?

A. $f(x) = 0.02(3,000)^x$

B. $f(x) = 0.98(3,000)^x$

C. $f(x) = 3,000(0.02)^x$

D. $f(x) = 3,000(0.98)^x$

Q3

ADVANCED MATH

$$g(x) = x^2 + 55$$

What is the minimum value of the given function?

- A. 0
- B. 55
- C. 110
- D. 3,025

Q4

ADVANCED MATH

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

- A. 0
- B. 4.9
- C. 7
- D. 9

Q5

ADVANCED MATH

$$px + 57 = x^2$$

The given equation relates the value of x and its corresponding value of px for the function p . What is the minimum value of the function p ?

- A. -3,249
- B. -57
- C. 57
- D. 3,249

Q6

ADVANCED MATH

The function f is defined by $fx = 2700.1^x$. What is the value of $f0$?

- A. 0
- B. 1
- C. 27
- D. 270

The function $f(x) = 206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. Which of the following is the best interpretation of “ $f(5)$ is approximately equal to 243” in this context?

- A. The value of the bank account is estimated to be approximately 5 dollars greater in 1962 than in 1957.
- B. The value of the bank account is estimated to be approximately 243 dollars in 1962.
- C. The value, in dollars, of the bank account is estimated to be approximately 5 times greater in 1962 than in 1957.
- D. The value of the bank account is estimated to increase by approximately 243 dollars every 5 years between 1957 and 1972.

Q8

ADVANCED MATH

x	fx
-1	10
0	14
1	20

For the quadratic function f , the table shows three values of x and their corresponding values of fx . Which equation defines f ?

- A. $fx = 3x^2 + 3x + 14$
- B. $fx = 5x^2 + x + 14$
- C. $fx = 9x^2 - x + 14$
- D. $fx = x^2 + 5x + 14$

Q9

ADVANCED MATH

$$gx = 11\frac{1}{12}x$$

If the given function g is graphed in the xy -plane, where $y = gx$, what is the y -intercept of the graph?

- A. 0, 11
- B. 0, 132
- C. 0, 1
- D. 0, 12

$$f(x) = 3,000(0.75)^x$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 3,000 in this context?

- A. The estimated percent decrease in the population for this species and area every 8 years after 2008
- B. The estimated percent decrease in the population for this species and area each year after 2008
- C. The estimated population for this species and area 8 years after 2008
- D. The estimated initial population for this species and area in 2008